The Complete Spectrum of the Fair Lamplighter Chain on a Finite Cycle

Source-local theorem package HCS-C341

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Abstract

We diagonalize the fair switch–walk–switch chain on binary lamps over every finite cycle. Walsh transformation turns switching into deletion projections; each nontrivial block is a direct sum of killed lazy paths. This yields the full characteristic polynomial, eigenbasis, multiplicities, sharp spectral gap and exact boundary cases. Finite enumeration is used only as a regression receipt. No arithmetic target or Route-B claim is made.

1 Frozen model

Write $C_n = \mathbb{Z}/n\mathbb{Z}$ and $\Omega_n = \mathbb{Z}_2^{C_n} \times C_n$. For $n \geq 3$ the base kernel Q_n stays with probability $1/2$ and moves to each neighbour with probability $1/4$. At $n = 2$ the two directed choices coalesce, so the other vertex has probability $1/2$; $Q_1 = (1)$. One chain step independently replaces the lamp at the departure site by a fair bit, applies Q_n, and independently replaces the arrival lamp by a fair bit. Replacement, not deterministic toggling, is part of the definition.

For $A \subseteq C_n$ define the lamp character $\chi_A(\eta) = (-1)^{\sum_{x\in A}\eta(x)}$ and the position projection $D_A(x,x) = \mathbf{1}_{\{x\notin A\}}$. If R_A denotes the multiset of positive lengths of the cyclic runs in $C_n \setminus A$, put

$$c_{n,k} = \frac{1+\cos(2\pi k/n)}{2}, \qquad p_{\ell,j} = \frac{1}{2} + \frac{1}{2}\cos\frac{\pi j}{\ell+1}.$$

Theorem 1 (Full finite-cycle spectrum)**.** *The uniform law on Ω_n is the unique reversible stationary law and*

$$P_n \cong \bigoplus_{A\subseteq C_n} D_A Q_n D_A. \tag{1}$$

The empty-support block has eigenvalues $c_{n,k}$, $0 \leq k < n$. Every $A \neq \varnothing$ block has $|A|$ zero eigenvalues and, for each $\ell \in R_A$, the simple eigenvalues $p_{\ell,j}$, $1 \leq j \leq \ell$. Walsh characters times deleted-site deltas and path sines form a complete orthogonal eigenbasis. Consequently

$$\det(zI - P_n) = \det(zI - Q_n) \prod_{\varnothing\neq A\subseteq C_n} z^{|A|} \prod_{\ell\in R_A}\prod_{j=1}^{\ell}(z - p_{\ell,j}). \tag{2}$$

For $n \geq 3$ the largest nonconstant eigenvalue has multiplicity n and

$$\lambda_{\mathrm{top}} = \frac{1+\cos(\pi/n)}{2}, \qquad \mathrm{gap}(P_n) = \frac{1-\cos(\pi/n)}{2}. \tag{3}$$

For every mean-zero f, $\|P_n^t f\|_2 \leq \lambda_{\mathrm{top}}^t \|f\|_2$, and equality is attained. At $n = 2$ the gap is $1/2$; at $n = 1$ the spectrum is $\{1, 0\}$ and the gap is one.

2 Walsh blocks

Let S be one fair switch. It is conditional expectation over the current lamp, hence an orthogonal projection in uniform L^2. The symmetric kernel Q_n is self-adjoint, so $P_n = SQ_nS$ is self-adjoint and preserves constants. A positive-probability path can visit every site and choose prescribed switch outcomes; the finite chain is irreducible, proving uniqueness.

On $\chi_A(\eta)g(x)$, fair averaging gives

$$S(\chi_A g) = \chi_A D_A g :$$

the character is unchanged when $x \notin A$ and has mean zero when $x \in A$. The position move leaves the lamp character fixed. Thus switch, move and switch acts as $D_A Q_n D_A$, proving (1).

3 Path diagonalization and completeness

For $A = \varnothing$, cyclic Fourier vectors give $c_{n,k}$. If $A \neq \varnothing$, the coordinates in A are killed and supply $|A|$ zero vectors. Every remaining run of length ℓ carries the tridiagonal matrix with diagonal 1/2 and neighbouring entries 1/4. With Dirichlet endpoints,

$$u_j(r) = \sin \frac{\pi j r}{\ell + 1}, \qquad 1 \leq r \leq \ell,$$

satisfies

$$\tfrac{1}{4} u_j(r-1) + \tfrac{1}{2} u_j(r) + \tfrac{1}{4} u_j(r+1) = p_{\ell,j} u_j(r).$$

Discrete sine orthogonality gives ℓ vectors on each run. Together with the deleted deltas there are $|A| + \sum_{\ell \in R_A} \ell = n$ vectors in sector A; distinct sectors are Walsh-orthogonal. The basis and product formula (2) follow.

4 The sharp mode and small cycles

All eigenvalues are nonnegative. The empty sector's largest value below one is $(1 + \cos(2\pi/n))/2$. A nonempty support leaves runs of length at most $n - 1$, while $(1 + \cos(\pi/(\ell + 1)))/2$ increases strictly with ℓ. The maximum below one is therefore (3), attained exactly by singleton supports. Each corresponding path top eigenvalue is simple, so its multiplicity is exactly n. The spectral theorem gives the stated contraction and an eigenfunction gives equality.

For $n = 2$, the empty block is $\{1, 0\}$, each singleton-support block is $\{1/2, 0\}$, and the full-support block is zero. For $n = 1$, the position is fixed and the fair lamp replacement has spectrum $\{1, 0\}$. These computations also show why directed-neighbour multiplicities must be coalesced before forming Q_2.

Round-one proof strengthening. The previous paragraphs add the complete eigenbasis, the uniqueness of the top-mode owner, its multiplicity, sharpness, and both degenerate-cycle cases.

5 Exact evidence and claim firewall

The canonical evidence enumerates all 2046 Walsh blocks for $1 \leq n \leq 10$. It records cyclic runs and rational monic continuant factors, while full exact kernels through $n = 6$ independently check stochasticity, symmetry and two spectral traces. A producer-independent checker reconstructs every coordinate; separate SymPy, isolated byte-replay and repaired-hash mutation lanes test the implementation. These finite checks do not prove the universal theorem.

The nearest retained systems are independent hypercube flips (C171), random transpositions (C183), chamber walks (C192), and Wilson cycle popping (C338). None owns a moving lamp field or (1). The finite self-adjoint contraction is only a source-side formal hint. We claim no arithmetic local data, target Euler factor, root number, automorphy, target divisor, target zero match, Hilbert–Pólya operator, or Route-B readiness.

Round-two release strengthening. This revision binds the finite receipt, collision boundary, sources and Route-A scope to the proved theorem.

6 Source lineage

The general switch–walk–switch/percolation correspondence is due to Lehner, Neuhauser and Woess, *Math. Ann.* 342 (2008), 69–89, doi:10.1007/s00208-008-0222-7. Graph-level eigenspaces were developed by Lehner, *Proc. AMS* 137 (2009), 2631–2637, doi:10.1090/S0002-9939-09-09869-4. The present finite-cycle theorem is a self-contained specialization and makes no priority claim for that general theory.